

MedIntel – AI Medical Research

Project Initiation Document (PID)

Project Title

MedIntel – AI Medical Research Assistant

Version: 1.0

Project Type: Generative AI Web Application

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1. Project Overview

MedIntel is an AI-powered Medical Research Assistant designed to simplify the analysis of medical research papers using Generative Artificial Intelligence and Retrieval-Augmented Generation (RAG). The application enables users to upload medical documents in PDF format and interact with them through an intelligent conversational interface capable of providing context-aware responses and concise summaries.

The system integrates modern web technologies with Large Language Models to create an efficient research assistant that improves accessibility to complex medical literature while ensuring responses remain grounded in the uploaded documents.

2. Background

Medical researchers, healthcare professionals, and students frequently work with lengthy research papers that require considerable effort to understand and analyze. Traditional keyword-based searching often fails to identify semantically relevant information, leading to inefficient research workflows.

Recent advancements in Large Language Models and Retrieval-Augmented Generation provide an opportunity to build intelligent systems capable of understanding document context and delivering precise answers based on uploaded content.

This project aims to leverage these technologies to create an AI assistant specifically designed for medical research.

3. Problem Statement

Medical research documents are often extensive, highly technical, and time-consuming to analyze manually.

Current challenges include:

- Difficulty locating relevant information quickly.
- Time-consuming manual reading.
- Limited semantic understanding in traditional search systems.
- Generic AI chatbots producing hallucinated or unsupported responses.
- Lack of document-specific conversational interfaces.

These limitations reduce research productivity and increase the effort required to extract meaningful insights.

4. Proposed Solution

Develop an AI-powered web application that combines Retrieval-Augmented Generation with Google Gemini to provide document-grounded responses.

The proposed solution will:

- Allow users to securely upload medical PDFs.
 - Process and index uploaded documents.
 - Generate vector embeddings for semantic retrieval.
 - Answer questions using only uploaded document content.
 - Produce concise document summaries.
 - Maintain user conversation history.
 - Provide a modern and responsive web interface.
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5. Project Objectives

The primary objectives of MedIntel are:

- Develop a secure AI-powered medical research assistant.
- Implement Retrieval-Augmented Generation.

- Enable intelligent PDF-based question answering.
 - Generate AI-powered summaries.
 - Maintain conversation history.
 - Deploy the application to cloud platforms.
 - Deliver a responsive and user-friendly interface.
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6. Scope of the Project

In Scope

- User Registration
- User Login
- JWT Authentication
- Medical PDF Upload
- PDF Text Extraction
- Text Chunking
- Embedding Generation
- Vector Database Storage
- Semantic Retrieval
- AI Chat
- AI Summary Generation
- Conversation History
- Cloud Deployment

Out of Scope

- Medical diagnosis
- Image-based medical analysis
- Multi-document comparison
- Voice interaction

- Mobile application
 - Electronic Health Record integration
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7. Business Objectives

The proposed application aims to:

- Improve medical literature accessibility.
 - Reduce document analysis time.
 - Enhance research productivity.
 - Demonstrate practical implementation of Generative AI.
 - Showcase Retrieval-Augmented Generation in healthcare applications.
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8. Functional Requirements

ID	Requirement
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FR-01	User Registration
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FR-02	User Login
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FR-03	Upload Medical PDF
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FR-04	Generate Summary
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FR-05	AI Chat
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FR-06	Semantic Search
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FR-07	Conversation History
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FR-08	Secure Authentication
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9. Non-Functional Requirements

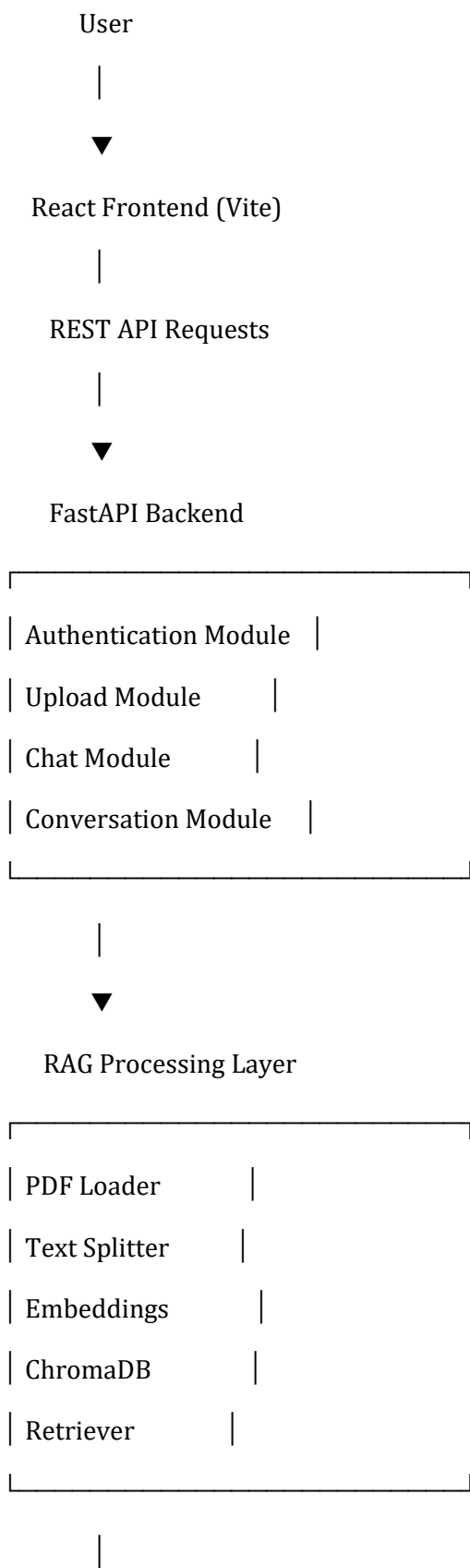
Category	Requirement
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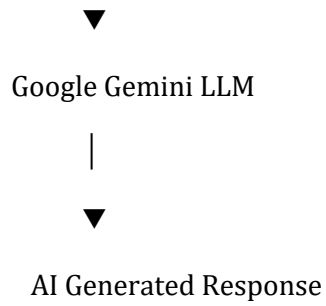
Category	Requirement
Performance	Fast document indexing
Security	JWT Authentication
Availability	Cloud Deployment
Reliability	Error Handling
Maintainability	Modular Architecture
Scalability	Expandable AI Pipeline
Usability	Responsive UI

10. Technology Stack

Component	Technology
Frontend	React
Build Tool	Vite
Styling	Tailwind CSS
Backend	FastAPI
Language	Python
AI Model	Google Gemini
Framework	LangChain
Vector Database	ChromaDB
Database	SQLite
Authentication	JWT
Deployment	Render & Vercel

11. Proposed System Architecture





12. Expected Workflow

1. User Registration
 2. User Login
 3. Upload Medical PDF
 4. Extract Document Text
 5. Generate Text Chunks
 6. Create Embeddings
 7. Store in ChromaDB
 8. Ask Questions
 9. Retrieve Relevant Context
 10. Generate AI Response
 11. Display Response
 12. Store Conversation
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13. Risks and Mitigation

Risk	Mitigation Strategy
Large PDF processing time	Optimize chunk size and retrieval
AI hallucinations	Use Retrieval-Augmented Generation
Deployment failures	Deploy frontend and backend independently

Risk	Mitigation Strategy
API limitations	Secure API management and fallback handling
Authentication vulnerabilities	JWT-based secure authentication

14. Expected Deliverables

The project is expected to deliver:

- Fully functional web application
 - AI-powered chatbot
 - Medical PDF processing
 - Document summarization
 - Semantic search
 - User authentication
 - Conversation history
 - Cloud deployment
 - Project documentation
 - Source code repository
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15. Success Criteria

The project will be considered successful if it meets the following criteria:

- Users can register and log in securely.
- Medical PDFs can be uploaded successfully.
- Uploaded documents are indexed correctly.
- AI generates accurate document summaries.
- Users receive context-aware answers.
- Conversation history is maintained.
- Frontend and backend are successfully deployed.

- System demonstrates Retrieval-Augmented Generation effectively.
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16. Assumptions

The project assumes that:

- Users upload valid medical research PDFs.
 - Internet connectivity is available for AI inference.
 - Google Gemini API remains accessible.
 - Cloud deployment services are available.
 - The application is intended for research assistance and educational purposes rather than clinical decision-making.
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17. Conclusion

The MedIntel project aims to demonstrate the practical integration of Generative AI and Retrieval-Augmented Generation in the healthcare research domain. By combining secure authentication, intelligent document processing, semantic retrieval, and conversational AI, the application seeks to provide an efficient platform for interacting with medical research papers. The proposed solution emphasizes accuracy, scalability, maintainability, and user experience while serving as a real-world implementation of modern AI technologies.